

Week 42 / Fair results from a bag / Grades 2-5

Use one unchanged bag containing both colors. Choose each draw independently without looking; each counter has the same chance. Return it and mix after every draw. A rule sees only the color word, in order.

Problem 1: Make a three-draw rule that gives a square, circle, or triangle with equal chances for every fixed bag containing both colors. Skip RRR and BBB; use every other word.



			output _____
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			output _____
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			output _____
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			output _____
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			output _____
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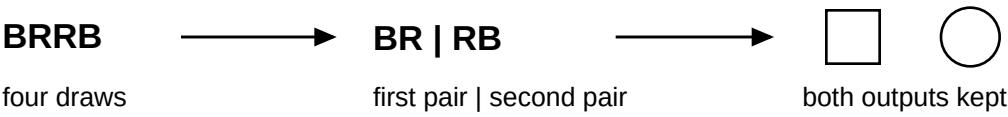
			output _____
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			output _____
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			output _____
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Week 42 / Fair results from a bag / Grades 4-5

For each pair: BR gives a square, RB a circle, and RR or BB gives nothing. Make four draws from one fixed bag; use the first pair and the second pair. Keep both outputs already made.



Problem 2: If both pairs were skipped, can you use them to make one extra fair output? Design a rule. Compare output totals on all sixteen equally likely stories from a one-red, one-blue bag. Would it also work for three reds and one blue?

RR RR outputs: _____	RR RB outputs: _____	RR BR outputs: _____	RR BB outputs: _____
RB RR outputs: _____	RB RB outputs: _____	RB BR outputs: _____	RB BB outputs: _____
BR RR outputs: _____	BR RB outputs: _____	BR BR outputs: _____	BR BB outputs: _____
BB RR outputs: _____	BB RB outputs: _____	BB BR outputs: _____	BB BB outputs: _____

		
Basic		
Recycled		

Problem 3: Make two cups of four equal-chance tickets, each showing a left shape and a right shape. At each position, square and circle must have equal chances. One cup must give only two possible pairs; the other must give all four. Does fairness at each position make the whole pair fair?

Cup 1

left	right
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left	right
------	-------

left	right
------	-------

left	right
------	-------

Cup 2

left	right
------	-------

left	right
------	-------

left	right
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left	right
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